数理方程

第二周 20260306

主要内容 波动方程 〈 齐次：行波解（d'Alembert解）
非齐次：齐次化原理

1) $\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f$ 波动方程 $L = \frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2}$

2) $\frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = f$ 热传导 $L = \frac{\partial}{\partial t} - k \frac{\partial^2}{\partial x^2}$

3) $\nabla^2 u = f$ 泊松方程 $L = \nabla^2$

共性：线性方程 $L(u) = f$（u：未知函数，f：已知函数）

L为线性算符：$L(C_1 u_1 + C_2 u_2) = C_1 L(u_1) + C_2 L(u_2)$（$C_1$、$C_2$：常数）

例：$\left(\frac{\partial}{\partial t} - k \frac{\partial^2}{\partial x^2}\right)(C_1 u_1 + C_2 u_2) = C_1 \left(\frac{\partial}{\partial t} - k \frac{\partial^2}{\partial x^2}\right) u_1 + C_2 \left(\frac{\partial}{\partial t} - k \frac{\partial^2}{\partial x^2}\right) u_2$（线性）

非线性算符例：$M(u) = u \frac{\partial u}{\partial x} \Rightarrow (C_1 u_1 + C_2 u_2) \frac{\partial}{\partial x}(C_1 u_1 + C_2 u_2) \neq C_1 u_1 \frac{\partial u_1}{\partial x} + C_2 u_2 \frac{\partial u_2}{\partial x}$

$\Rightarrow M$为非线性算符

〈 $f = 0$ 齐次方程
$f \neq 0$ 非齐次方程

例：$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$不是线性方程，解的叠加不再是解

解的叠加原理

性质1：$L(u_1) = 0$，$L(u_2) = 0 \Rightarrow L(C_1 u_1 + C_2 u_2) = 0$

齐次方程的解作线性组合（叠加）仍然是解

性质2：$L(u_1) = 0$，$L(u_2) = f \Rightarrow L(u_1 + u_2) = f$

齐次方程解 + 非齐次方程解 = 非齐次方程解

性质3：$L(u_1) = f_1$，$L(u_2) = f_2 \Rightarrow L(C_1 u_1 + C_2 u_2) = C_1 f_1 + C_2 f_2$

非齐次项作线性组合，解也作相应线性组合

$u_1, u_2 \cdots u_n$: $L(u_1)=f_1, \cdots L(u_n)=f_n$

$\Rightarrow L(C_1u_1+\cdots+C_nu_n) = C_1f_1+\cdots+C_nf_n$

ODE $\frac{d^2y}{dt^2}=-g \Rightarrow y(t) = -\frac{g}{2}t^2+A+Bt$ 通解

任意常数 特解 → 代入初始条件

PDE:

1阶PDE $u_t+au_x=0$, $-\infty<x<\infty$, $u|_{t=0}=\phi(x)$

新变量 $\begin{cases}\eta = x-at \\ \xi 与 \eta 独立. 例: \xi = x, \ \xi = x+at \cdots.\end{cases}$

$\begin{cases}\eta(x,t), \\ \xi(x,t).\end{cases} \Rightarrow (\eta_t u_\eta + \xi_t u_\xi) + a(\eta_x u_\eta + \xi_x u_\xi) = 0$

$\Rightarrow (\eta_t+a\eta_x)u_\eta + (\xi_t+a\xi_x)u_\xi = 0$

$\eta = x-at \Rightarrow -a+a=0$; $\xi_t+a\xi_x \neq 0$

$\Downarrow$

$u_\xi = 0$

$\Downarrow$

$u = f(\eta) = f(x-at)$

任意函数 通解

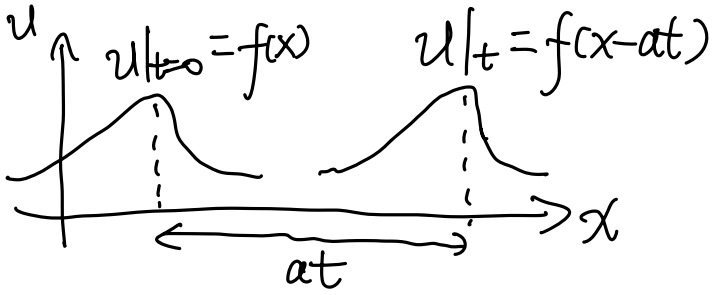
通解 $\begin{cases}\text{ODE: 含任意常数} \\ \text{PDE: 含任意函数}\end{cases}$

IC $u|_{t=0}=\phi(x) \Rightarrow f(x)=\phi(x)$

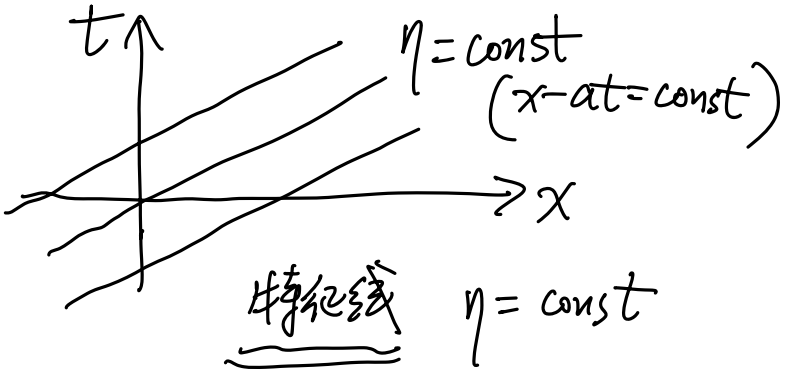
$\Rightarrow u(x,t)=\phi(x-at)$

$u=f(\eta) \Rightarrow$ 若 $\eta=\text{const}$, 则 $u=\text{const}$

即 $x-at=\text{const}$



波速 a



特征线 $\eta=\text{const}$

沿波前行，波形不变

函数 u 在特征线上 = const

特征线：波传播的路径 $(\eta=x-at=\text{const})$

特征坐标：简化 PDE 的坐标 $(\eta=x-at, \xi=\dots)$

二阶PDE. 行波解（d'Alembert解）

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2 \dfrac{\partial^2 u}{\partial x^2} = 0, & -\infty < x < \infty, \quad t > 0 \\ u|_{t=0} = \phi(x), \quad \dfrac{\partial u}{\partial t}\Big|_{t=0} = \psi(x) \end{cases} \Rightarrow u(x,t) = ?$$

求PDE通解：

$$\left(\frac{\partial}{\partial t} + a\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} - a\frac{\partial}{\partial x}\right)u = 0 \xrightarrow{\text{希望}} \begin{cases} \dfrac{\partial}{\partial t} + a\dfrac{\partial}{\partial x} \sim \dfrac{\partial}{\partial \eta} \\ \dfrac{\partial}{\partial t} - a\dfrac{\partial}{\partial x} \sim \dfrac{\partial}{\partial \xi} \end{cases}$$

$\xi = x - at$, $\eta = x + at$ $\Longleftrightarrow$ $x = \dfrac{\xi+\eta}{2}$, $t = \dfrac{-\xi+\eta}{2a}$

$$\frac{\partial}{\partial \xi} = \frac{\partial t}{\partial \xi}\frac{\partial}{\partial t} + \frac{\partial x}{\partial \xi}\frac{\partial}{\partial x} = \frac{1}{2a}\left(-\frac{\partial}{\partial t} + a\frac{\partial}{\partial x}\right), \qquad \frac{\partial}{\partial \eta} = \frac{\partial t}{\partial \eta}\frac{\partial}{\partial t} + \frac{\partial x}{\partial \eta}\frac{\partial}{\partial x} = \frac{1}{2a}\left(\frac{\partial}{\partial t} + a\frac{\partial}{\partial x}\right)$$

$$\Rightarrow \boxed{\frac{\partial^2 u}{\partial \xi \partial \eta} = 0} \Rightarrow \frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial \xi}\right) = 0 \Rightarrow \frac{\partial u}{\partial \xi} = A(\xi)$$

⇓

$$u = \int_{\xi_0}^{\xi} A(\xi')\,d\xi' + g(\eta)$$

$\int_{\xi_0}^{\xi} A(\xi')\,d\xi' = f(\xi)$；$g(\eta)$：“积分常数”

$$= f(\xi) + g(\eta)$$

⇓

$$u(x,t) = f(x-at) + g(x+at)$$

含任意函数 f, g 通解

$f(x-at)$：右行波　　$g(x+at)$：左行波

没有 $\frac{\partial^2 u}{\partial \xi^2}$ 和 $\frac{\partial^2 u}{\partial \eta^2}$ 项。更细致的推导是：设 $\xi = At + Bx$, $\eta = Ct + Dx$, 其中 A, B, C, D 为常数 $\Rightarrow \frac{\partial^2}{\partial t^2} - a^2\frac{\partial^2}{\partial x^2}$

$$= \left(\frac{\partial \xi}{\partial t}\frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial t}\frac{\partial}{\partial \eta}\right)^2 - a^2\left(\frac{\partial \xi}{\partial x}\frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial x}\frac{\partial}{\partial \eta}\right)^2$$

$$= (A^2 - a^2B^2)\frac{\partial^2}{\partial \xi^2} + (C^2 - a^2D^2)\frac{\partial^2}{\partial \eta^2} + 2(AC - a^2BD)\frac{\partial^2}{\partial \xi \partial \eta}$$

取 $A^2 = a^2B^2$, $C^2 = a^2D^2$ 可使前两项消失.

最简取法为 $B = D = 1$, $A = -a$, $C = a$,

此即我们的 (ξ, η) 取法.



右行波 波速 a

$u|_{t=0} = f(x)+g(x)=\phi(x)$, $\left.\frac{\partial u}{\partial t}\right|_{t=0} = -af'(x-at)+ag'(x+at)\Big|_{t=0}$

$= -af'(x)+ag'(x)=\psi(x)$

$\Downarrow$

$-f(x)+g(x)=\frac{1}{a}\int_0^x \psi(\xi)d\xi + C$

$\Rightarrow f(x)=\frac{1}{2}\left[\phi(x)-\frac{1}{a}\int_0^x \psi(\xi)d\xi - C\right]$

$g(x)=\frac{1}{2}\left[\phi(x)+\frac{1}{a}\int_0^x \psi(\xi)d\xi + C\right]$ $\Rightarrow$ $f(x-at)=\frac{1}{2}\left[\phi(x-at)-\frac{1}{a}\int_0^{x-at}\psi(\xi)d\xi - C\right]$

$g(x+at)=\frac{1}{2}\left[\phi(x+at)+\frac{1}{a}\int_0^{x+at}\psi(\xi)d\xi + C\right]$

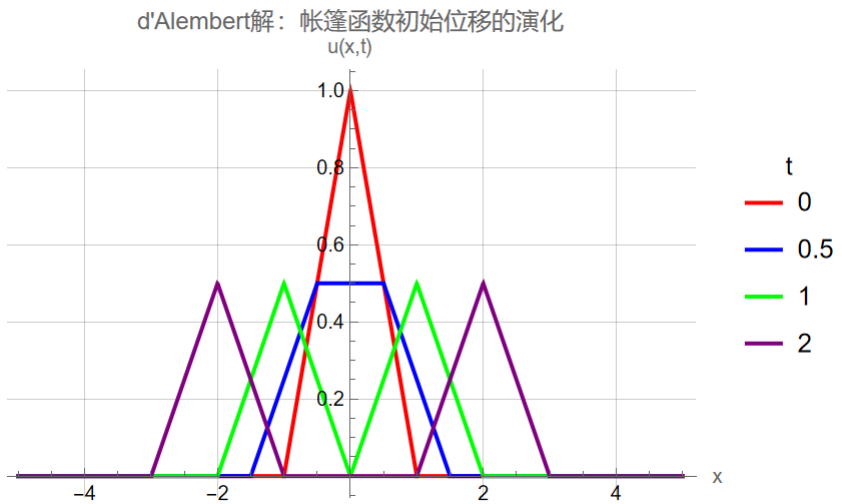
$\Rightarrow u(x,t)=f(x-at)+g(x+at)=\frac{1}{2}\left[\phi(x-at)+\phi(x+at)\right]+\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)\,d\xi$

行波解，d'Alembert解

1746年

例：初位移为$\phi(x)$ = tent function

初速$\psi(x)=0$

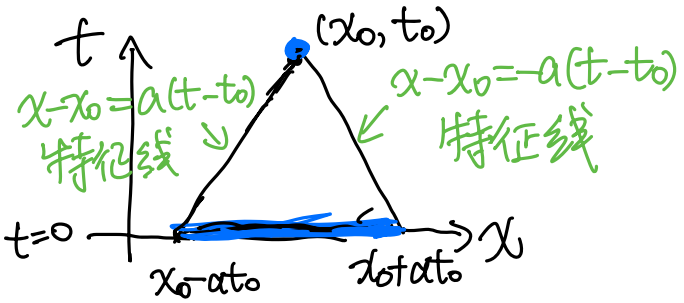


问：$\begin{cases} \frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial^2 u}{\partial x^2}=0 & -\infty<x<\infty,\ t>\tau \\ u|_{t=\tau}=\phi(x),\ \left.\frac{\partial u}{\partial t}\right|_{t=\tau}=\psi(x) \end{cases}$

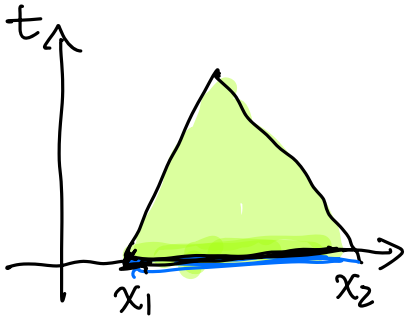
$t'=t-\tau$

$\Rightarrow u(x,t)=\frac{1}{2}\left[\phi(x-a(t-\tau))+\phi(x+a(t-\tau))\right]+\frac{1}{2a}\int_{x-a(t-\tau)}^{x+a(t-\tau)}\psi(\xi)d\xi$

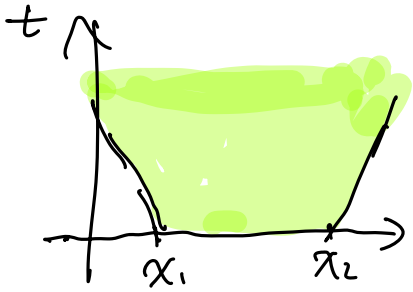
留意，稍后用到



(x_0,t_0)的依赖区间



$[x_1,x_2]$的决定区域



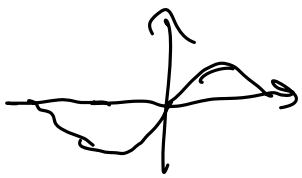
$[x_1,x_2]$的影响区域

半无界问题：

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2}-a^2\dfrac{\partial^2 u}{\partial x^2}=0, & x>0 \quad t>0 \\ u|_{x=0}=0 \\ u|_{t=0}=\phi(x),\ \dfrac{\partial u}{\partial t}\Big|_{t=0}=\psi(x) & (x>0) \end{cases}$$



奇延拓 $\Phi(x)=\begin{cases}\phi(x) & x>0\\ -\phi(-x) & x<0\end{cases}$ $\Psi(x)=\begin{cases}\psi(x), & x>0\\ -\psi(-x) & x<0\end{cases}$



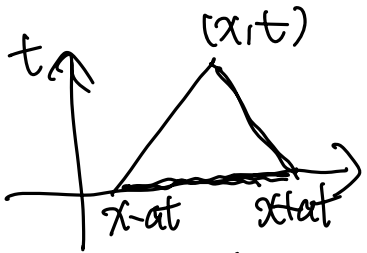
$u(x,t)=\frac{1}{2}[\Phi(x-at)+\Phi(x+at)]+\frac{1}{2a}\int_{x-at}^{x+at}\Psi(\xi)d\xi$ 在$(-\infty,+\infty)$上满足PDE

由于奇延拓，此解满足边界条件 $u|_{x=0}=0$

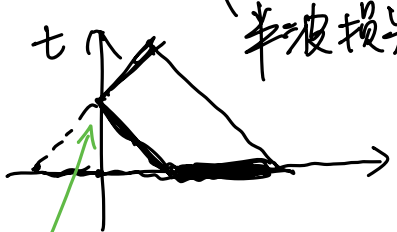
但由于$x<0$是虚拟的，最后结果应由$x>0$区间的量来表达：

$$u(x,t)=\begin{cases}\frac{1}{2}[\phi(x+at)+\phi(x-at)]+\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi, & t\le\frac{x}{a}\\ \frac{1}{2}[\phi(x+at)-\phi(at-x)]+\frac{1}{2a}\int_{at-x}^{x+at}\psi(\xi)d\xi, & t>\frac{x}{a}\end{cases}$$

半波损失



$t\le\frac{x}{a}$



边界反射

$t>\frac{x}{a}$

问：如果$u|_{x=0}=0$换为$\dfrac{\partial u}{\partial x}\Big|_{x=0}=0$，如何解？

作业

非齐次方程的齐次化原理 Duhamel Principle 1797-1872 冲量定理

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = a^2 \dfrac{\partial^2 u}{\partial x^2} + f(x,t), \quad -\infty < x < \infty, \quad t > 0 \\ u|_{t=0} = 0, \quad \dfrac{\partial u}{\partial t}\Big|_{t=0} = 0 \end{cases}$$

若初始条件为 $u|_{t=0} = \phi(x)$, $\dfrac{\partial u}{\partial t}\Big|_{t=0} = \psi(x)$, 只需将问题拆成①②两部分求解.

① $\begin{cases} \dfrac{\partial^2 u_1}{\partial t^2} = a^2 \dfrac{\partial^2 u_1}{\partial x^2} \\ u_1|_{t=0} = \phi(x), \ \dfrac{\partial u_1}{\partial t}\Big|_{t=0} = \psi(x) \end{cases}$

② $\begin{cases} \dfrac{\partial^2 u_2}{\partial t^2} = a^2 \dfrac{\partial^2 u_2}{\partial x^2} + f \\ u_2|_{t=0} = \dfrac{\partial u_2}{\partial t}\Big|_{t=0} = 0 \end{cases}$

已经会解①, 只用解②

$\Downarrow$

$u = u_1 + u_2$

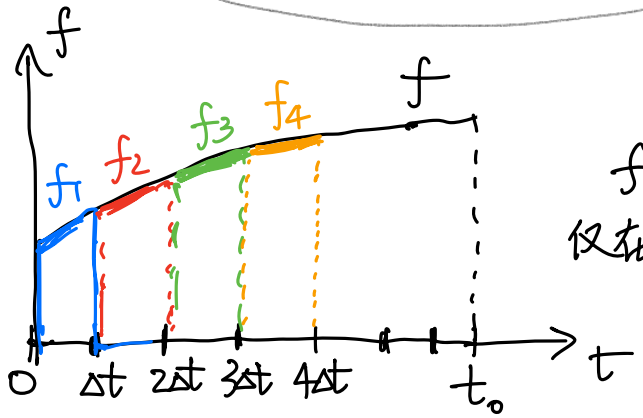
1) 分割

不想分割 $[0,\infty)$, 所以取某个 $t_0 > 0$

在 $[0, t_0]$ 区间里解方程

将 $[0, t_0]$ 分成 N 份

$N\Delta t = t_0$



f_n 脉冲力, 仅在小区间内非零

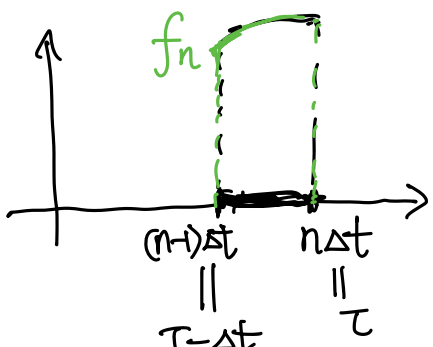
$$f_1(x,t) + f_2(x,t) + \cdots + f_N(x,t) = f(x,t)$$

$$\begin{cases} \dfrac{\partial^2 u_n}{\partial t^2} - a^2 \dfrac{\partial^2 u_n}{\partial x^2} = f_n(x,t), \quad -\infty < x < \infty, \ t > 0 \\ u_n|_{t=0} = \dfrac{\partial u_n}{\partial t}\Big|_{t=0} = 0 \end{cases} \Rightarrow u(x,t) = u_1(x,t) + \cdots + u_N(x,t)$$

$\Downarrow$ 叠加原理

2) 求解脉冲力问题 (关键)

$$f_n|_{t < \tau - \Delta t} = 0$$



在小区间内, f_n 近似为常数

$f_n(x,t) \approx f(x,\tau)$

$t \in [\tau - \Delta t, \tau]$

$$\Rightarrow u_n|_{t \le \tau - \Delta t} = 0, \quad \frac{\partial u_n}{\partial t}\Big|_{t \le \tau - \Delta t} = 0$$

经过区间 $[\tau-\Delta t, \tau]$，外力 f 产生速度 $\left.\frac{\partial u_n}{\partial t}\right|_{t=\tau} = f(x,\tau)\Delta t$，

位移 $u_n|_{t=\tau} \sim (\Delta t)^2$ 可忽略

$$\frac{\partial^2 u_n(x,t)}{\partial t^2} = f(x,\tau) \Longrightarrow \left.\frac{\partial u_n}{\partial t}\right|_{t=\tau} - \underbrace{\left.\frac{\partial u_n}{\partial t}\right|_{t=\tau-\Delta t}}_{=0} = f(x,\tau)\Delta t$$

$[\tau-\Delta t, \tau]$ 内，$a^2\frac{\partial^2 u_n}{\partial x^2} \sim (\Delta t)^2$，可忽略

$t>\tau$：
$$\begin{cases} \frac{\partial^2 u_n}{\partial t^2} - a^2\frac{\partial^2 u_n}{\partial x^2} = 0 & \leftarrow \text{齐次方程} \\ u_n|_{t=\tau} = 0 \end{cases} \qquad \left.\frac{\partial u_n}{\partial t}\right|_{t=\tau} = f(x,\tau)\Delta t$$

↓ 除以 Δt，使右边不含 Δt.

$$\left.\frac{\partial (u_n/\Delta t)}{\partial t}\right|_{t=\tau} = f(x,\tau)$$

定义 $w(x,t;\tau) = \frac{u_n(x,t)}{\Delta t}$（$\tau = n\Delta t$）$\Rightarrow$

$$\begin{cases} \frac{\partial w}{\partial t} - a^2\frac{\partial^2 w}{\partial x^2} = 0 \\ w(x,t;\tau)|_{t=\tau} = 0, \quad \left.\frac{\partial w(x,t;\tau)}{\partial t}\right|_{t=\tau} = f(x,\tau) \end{cases}$$

3) $\Delta t \to 0$，求和 $\to$ 积分

所有小区间贡献相加 “厚积薄发”

$$u(x,t) = u_1(x,t) + u_2(x,t) + u_3(x,t) + \cdots$$
$$= w(x,t;\tau_1)\Delta t + w(x,t;\tau_2)\Delta t + w(x,t;\tau_3)\Delta t + \cdots$$

（$\tau_1 = \Delta t$，$\tau_2 = 2\Delta t$，$\tau_3 = 3\Delta t$）

$$\xrightarrow{\Delta t \to 0} \int_0^t w(x,t;\tau)\,d\tau$$

↖ 按定义是 $\int_0^{t_0}$，但 $\tau > t$ 时 $w=0$，可换成 $\int_0^t$

将非齐次方程问题化为齐次方程问题，用后者的解来表示前者的解，所以称为齐次化原理.

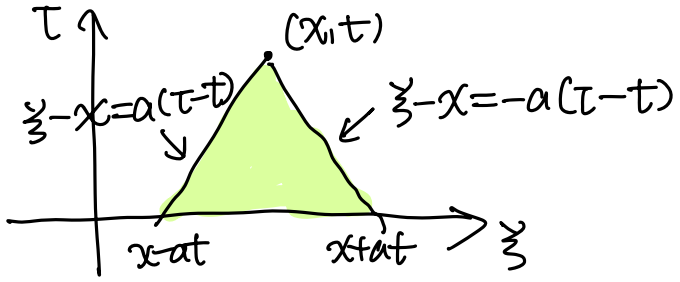
$w(x,t;\tau)$

τ时刻初速度

根据行波解: $w(x,t;\tau)=\frac{1}{2a}\int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi,\tau)d\xi$

$$\Rightarrow u(x,t)=\int_0^t w(x,t;\tau)d\tau=\frac{1}{2a}\int_0^t d\tau\int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi,\tau)d\xi$$

积分区域:



数学验证 $u(x,t)=\int_0^t w(x,t;\tau)d\tau$ 确实是解:

1) 初始条件 $u|_{t=0}=0$

$$\frac{\partial u}{\partial t}=\underbrace{w(x,t;t)}_{=0}+\int_0^t\frac{\partial w(x,t;\tau)}{\partial t}d\tau \Rightarrow \left.\frac{\partial u}{\partial t}\right|_{t=0}=0$$

变上限积分求导　对被积函数求导

2) 满足PDE

$$\frac{\partial^2 u}{\partial t^2}=\frac{\partial w(x,t;t)}{\partial t}+\int_0^t\frac{\partial^2 w(x,t;\tau)}{\partial t^2}d\tau.$$

$$=f(x,t)+a^2\int_0^t\frac{\partial^2 w}{\partial x^2}d\tau.$$

$$=f(x,t)+a^2\frac{\partial^2}{\partial x^2}\Big[\underbrace{\int_0^t w(x,t;\tau)d\tau}_{=u}\Big]$$

$$=f(x,t)+a^2\frac{\partial^2 u}{\partial x^2}$$

(时间二阶方程的) 齐次化原理:

L 为线性偏微分算子，其中对 t 的最高阶导数不超过 1 阶.

若 $w(\vec{x},t;\tau)$ 满足 $\begin{cases} \dfrac{\partial^2 w}{\partial t^2} = L(w), & t>\tau \\ w|_{t=\tau}=0, \ \dfrac{\partial w}{\partial t}\Big|_{t=\tau} = f(\vec{x},\tau) \end{cases}$

$\vec{x}$ 不限于一维，所以用元表示.

$\Rightarrow u(\vec{x},t;\tau)=\int_0^t w(\vec{x},t;\tau)\,d\tau$ 是相应非齐次方程之解:

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = L(u)+f(\vec{x},t), & t>0 \\ u|_{t=0} = \dfrac{\partial u}{\partial t}\Big|_{t=0} = 0. \end{cases}$$

例如, $L(u)=a\dfrac{\partial^2 u}{\partial x^2}+b\dfrac{\partial u}{\partial x}+cu+d\dfrac{\partial u}{\partial t}$ 符合条件.

本节总结 驭波而行 (行波解)
厚积薄发 (齐次化)